



VCU College of Engineering

CS 25-348 RenovationTracker

[Final Design Report]

Prepared for
Syed Khajamoinuddin
CoStar Group, Inc

By
Courtney Van
Jaidon Lightfoot
Jonathan Netala
Sterling Glasheen

Under the supervision of
Preetam Ghosh

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Executive Summary

RenovationTracker: Intelligent Renovation Detection and Visualization is a collaborative project with CoStar Group designed to enhance how property renovations are communicated in real estate listings. The project addresses the limitations of static property descriptions by introducing an AI-powered system that extracts, classifies, and presents renovation information in a clear, visual, and interactive format. This solution aims to support more informed decision-making for buyers and sellers by improving listing transparency and user engagement.

The system consists of two core technologies. First, a Natural Language Processing (NLP) model powered by OpenAI identifies renovation details from listing descriptions. Second, a computer vision model built with PyTorch classifies rooms in property images. These processes are run through AWS Lambda functions triggered by new data in an S3 bucket. The results are saved in JSON format, linking renovation text to specific rooms and images. This output is displayed through a React-based frontend that allows users to browse listings and see classified photos alongside summarized renovation details.

Key deliverables include a working end-to-end system that:

- Automatically processes scraped property listings and identifies renovations and rooms.
- Displays this information in an interactive, user-friendly interface.
- Operates on a scalable cloud-based architecture using AWS, OpenAI, and React.

Supporting tools include a web scraper that collects listings from homes.com, an S3-based pipeline for storing and triggering data processing, and a visual frontend that clearly presents the processed results. The backend is designed to be modular, supporting future expansion into additional data sources or models.

This project supports CoStar Group's goal of enhancing property data with advanced technologies. By combining automation, AI, and cloud computing, RenovationTracker provides a new way to surface high-value renovation information and sets a foundation for more dynamic, data-rich real estate listings.

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Section A. Problem Statement

In the real estate industry, clearly communicating the value of property renovations remains a major challenge. Listings typically include a set of static property images and a loosely written textual description, often composed by the seller or listing agent. These descriptions frequently lack detail and consistency, and the images are usually unstructured, leaving buyers to interpret the extent and quality of renovations on their own. This gap in presentation can lead to missed opportunities, lower engagement, and reduced perceived value. Sellers may struggle to effectively showcase high-cost upgrades, and buyers may skip listings that don't immediately communicate value.

This problem affects a wide range of stakeholders. Homeowners trying to sell, prospective buyers making decisions, agents writing listings at scale, and real estate platforms that host these listings all suffer from the limitations of the current process. Without accurate and visual representation of renovations, decision-making becomes slower and less reliable. The economic impact is significant, as listings that fail to stand out tend to stay on the market longer and may receive lower offers.

RenovationTracker was created to address this need by introducing a system that automatically detects and visualizes renovation data from real estate listings. Using artificial intelligence, the system extracts renovation-related text from property descriptions using a language model and identifies rooms in property images using a computer vision model. These outputs are connected through a cloud-based workflow and saved as structured data that links images to renovation details. The data is then displayed in an interactive interface that gives users a clearer and more engaging view of the property.

The project was developed in partnership with CoStar Group, a major player in the real estate analytics and listing industry. CoStar manages large volumes of residential and commercial listings, and this tool supports their ongoing efforts to integrate automation and artificial intelligence into their platform.

While other solutions exist in related areas, such as virtual staging or manual photo tagging tools, these are either costly, labor-intensive, or not scalable. RenovationTracker fills this gap by offering an automated, scalable system that requires no manual input and is capable of processing listings at scale. Similar research in natural language processing and computer vision has shown success in extracting structured data from real estate content, but few systems have integrated both technologies into a working tool focused specifically on renovations.

This project contributes to the advancement of real estate technology by reducing the manual effort required to communicate key features of a property. It improves listing quality, enhances transparency, and aligns with the growing trend of using AI to streamline high-volume content generation. It also positions CoStar to stay ahead in a competitive market by offering a more data-rich, user-friendly listing experience.

The main challenges lie in ensuring consistent model performance across diverse property styles and varying image quality, integrating the system into existing real estate workflows, and designing an intuitive interface that effectively communicates the extracted data. Despite these challenges, RenovationTracker demonstrates a practical and impactful approach to solving a well-defined problem in a high-traffic industry.

Section B. Engineering Design Requirements

B.1 Project Goals (Client Needs)

The RenovationTracker project aims to improve how property renovations are identified and communicated in the real estate industry. In collaboration with CoStar Group, the system is designed to automate the detection and presentation of renovations using artificial intelligence, enhancing user engagement and supporting better decision-making for buyers, sellers, and real estate professionals.

Key goals include:

- Enable accurate recognition of renovation details using natural language processing and computer vision.
- Provide dynamic visualization of renovation data, including potential side-by-side image comparisons.
- Deliver a user-friendly interface that presents information clearly and intuitively.
- Ensure the system architecture is scalable and adaptable to future needs.

B.2 Design Objectives

The design will:

- Achieve at least 90% accuracy in identifying and classifying renovations from text and image data.
- Display renovation information in an interactive frontend with a clean, intuitive layout.
- Integrate with real estate platforms through standardized APIs, including compatibility with CoStar infrastructure.
- Support scalability to handle large datasets, including up to 10,000 listings and associated images.
- Operate within a cloud-based architecture, maintaining reliable performance and efficient cost management.
- Be completed by the project deadline using available resources and within budget constraints.

B.3 Design Specifications and Constraints

- **System Integration:** Must support API-based interaction with platforms like CoStar for uploading or retrieving listing data.
- **Scalability:** The system must handle up to 10,000 properties, each with multiple images and descriptions, without significant degradation in performance.
- **Model Accuracy:** The natural language and computer vision models must maintain at least 90% accuracy in identifying renovations and room classifications across diverse property types.
- **Security and Privacy:** All user and property data must be handled in compliance with applicable data privacy laws, including GDPR, and stored securely using best practices.
- **Usability:** The user interface must be simple and intuitive enough for users with no technical background to access renovation information without guidance.
- **Interoperability:** The system must comply with REST API standards to ensure compatibility with existing web-based platforms.

B.4 Codes and Standards

- **ISO/IEC 27001 – Information Security Management**
Ensures that all sensitive property and user data is securely managed throughout the system.
- **ISO/IEC 25010 – Systems and Software Quality Requirements and Evaluation (SQuaRE)**
Used to guide quality assurance in areas such as functionality, reliability, and usability.
- **GDPR – General Data Protection Regulation**
Mandates proper handling of user and property data to ensure privacy compliance, particularly relevant for listings that include personal or sensitive content.
- **RFC 2616 – HTTP/1.1 Protocol (REST API Standards)**
Governs how the backend communicates over the web, ensuring standardized, reliable integration with external services like CoStar APIs.
- **ISO 9241-210 – Ergonomics of Human-System Interaction**
Guides the development of a usable and accessible user interface that supports non-technical users.

Section C. Scope of Work

The scope of the RenovationTracker project is to design, develop, and deploy an AI-powered system that automatically detects, classifies, and visualizes property renovations in real estate listings. The system combines Natural Language Processing (NLP) and computer vision technologies to extract renovation details from listing descriptions and classify room types from images. The final platform will present this data in an interactive interface, allowing users to better understand property improvements without relying on manually written descriptions or generic photo galleries.

The project focuses on delivering the following key objectives:

- **AI-powered renovation detection** using NLP to identify renovation-related information from property descriptions.
- **Automated room type classification** using computer vision to detect and label room types from property images.
- **Interactive renovation visualization** with side-by-side image comparisons and sliders to highlight changes.
- **System integration and testing** to combine all components into a unified platform that is tested for functionality, performance, and usability.

Project Boundaries and Responsibilities

The RenovationTracker team is responsible for designing, developing, and testing the core functionalities of the system, including text analysis, room classification, and interactive visualization. The team is also responsible for producing all academic deliverables required for course completion.

The following tasks fall outside of the team's scope:

- Development of tools or features unrelated to the RenovationTracker system.
- Server deployment logistics or costs, which are managed by CoStar.
- Collection of new data; the team will rely on datasets provided by CoStar or publicly available sources.
- Any physical hardware development or integration with external real estate platforms beyond software deliverables.

Development Methodology

The team is following an Agile development approach with iterative planning and implementation. Regular sprints, check-ins with faculty advisors and CoStar, and integration of feedback will guide the development process. This allows flexibility to adapt to technical challenges and evolving requirements while maintaining progress toward deliverables.

Timeline Boundaries and Risk Management

The project will be completed by the end of the academic year. Milestones are structured to meet academic deadlines for mid-term reports, final submissions, and the Capstone EXPO. Changes to the project scope will be discussed with CoStar and faculty advisors to ensure alignment and avoid scope creep.

Most of the work can be completed remotely, including model development, integration, testing, and documentation. Team members who regularly access campus will manage tasks that require on-site presence, such as poster printing and presentations.

To mitigate risks, the team will secure required software licenses, cloud access, and machine learning tools early. Regular coordination with CoStar ensures that server resources remain available for testing and deployment.

C.1 Deliverables

Primary deliverables include:

- An NLP model that extracts renovation data from listing descriptions.
- A computer vision model that classifies room types in property images.
- An integrated frontend interface that displays the output in an interactive format.
- A prototype visualization tool for side-by-side renovation comparisons.
- If time allows, a user feedback mechanism to confirm or correct renovation outputs.

Academic deliverables include:

- Team contract
- Project proposal
- Preliminary design report

- Fall presentation and poster
- Final design report
- Capstone EXPO poster and presentation

C.2 Milestones

The project is structured into the following milestones:

Milestone	Duration	Completion Date
Project proposal and requirements gathering	4 weeks	October 15, 2024
NLP model development	3 weeks	November 5, 2024
Image classification model development	3 weeks	November 26, 2024
Initial system integration	2 weeks	December 10, 2024
Preliminary system testing	2 weeks	January 10, 2025
Mid-term report and presentation	2 weeks	January 24, 2025
Renovation visualization tool development	3 weeks	February 14, 2025

Final system integration and refinement	3 weeks	March 20, 2025
Final testing and quality assurance	2 weeks	April 3, 2025
Final report and Capstone EXPO preparation	3 weeks	April 24, 2025
Capstone EXPO presentation	1 week	April 25, 2025

C.3 Resources

Software:

- NLP libraries
- Computer vision libraries (TensorFlow, PyTorch, Keras)
- IDEs
- GitHub for version control
- Cloud platforms for hosting and deployment (AWS, provided by CoStar)
- Optional: additional training resources via Google Cloud or Azure if needed

Hardware:

- Personal laptops and PCs for local development
- Cloud servers (provided by CoStar) for hosting and model execution

Data:

- Property descriptions and images (provided by CoStar)
- Pretrained models from open-source libraries

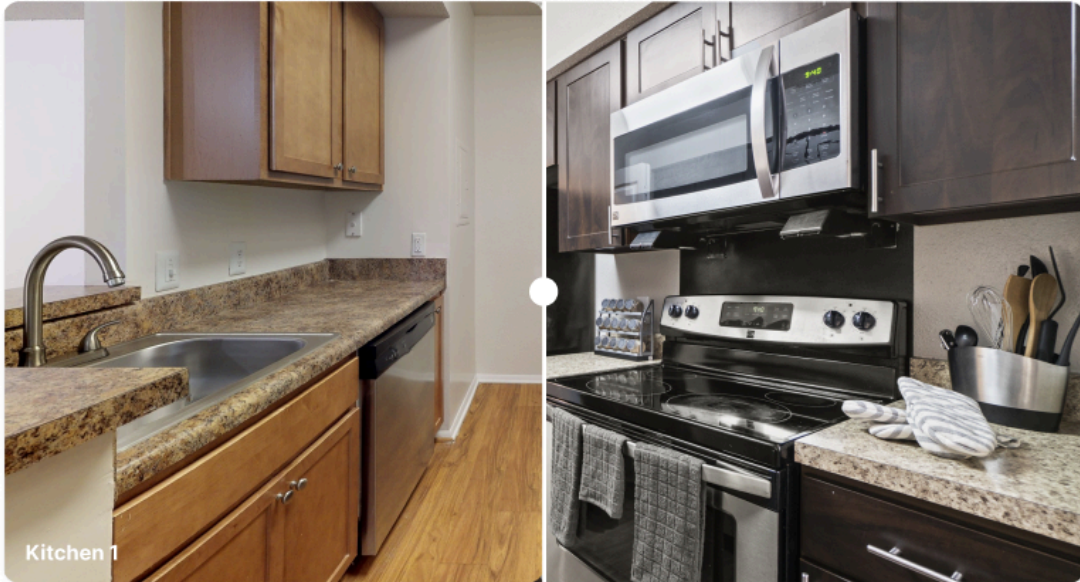
Additional Tools:

- Collaboration: Zoom
- Documentation: Google Docs, Microsoft Word

Section D. Concept Generation

Initial comparison prototype: Courtney 10/11/24

Renovation Comparison



< ● ○ ○ >

Kitchen

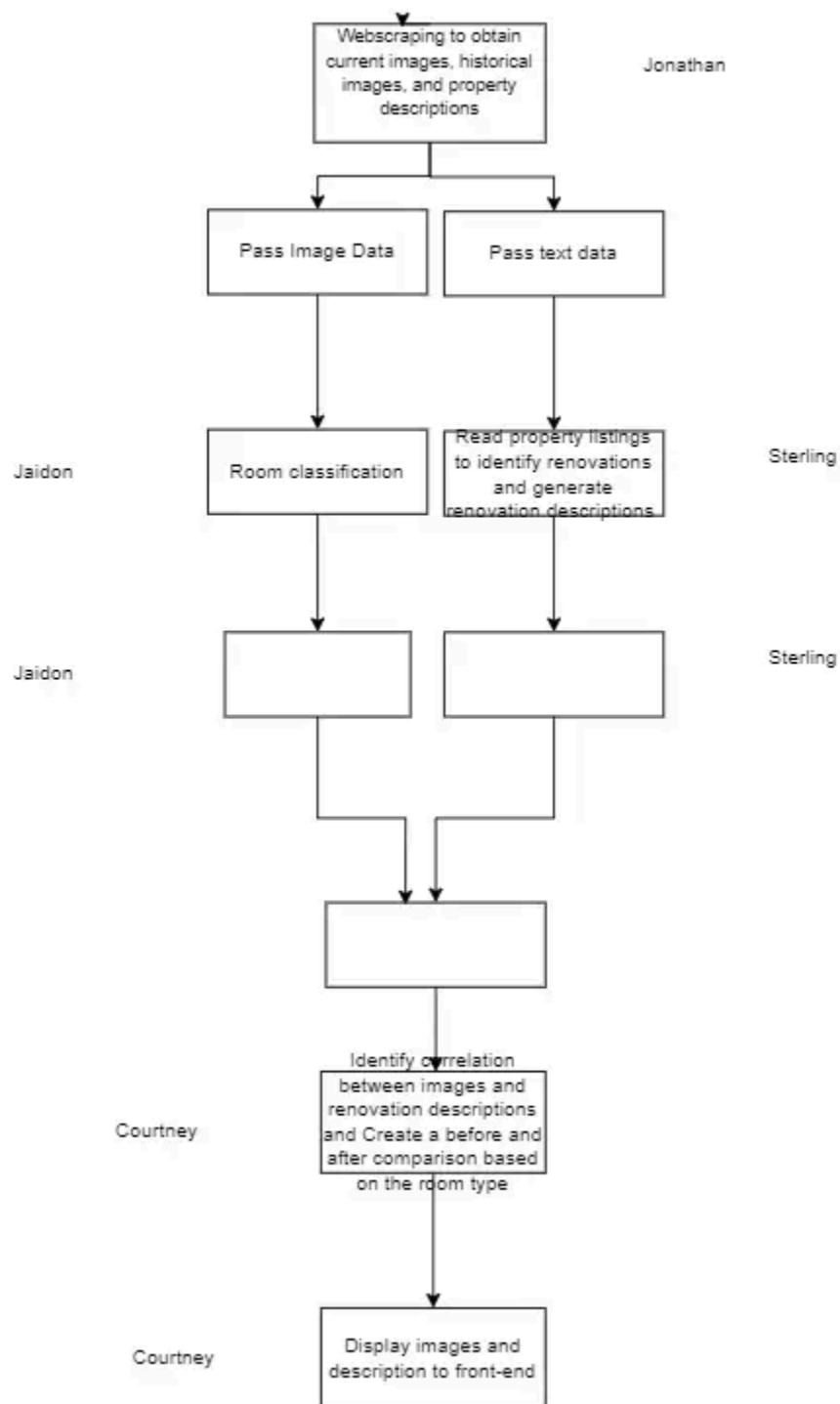
Bedroom

Exterior

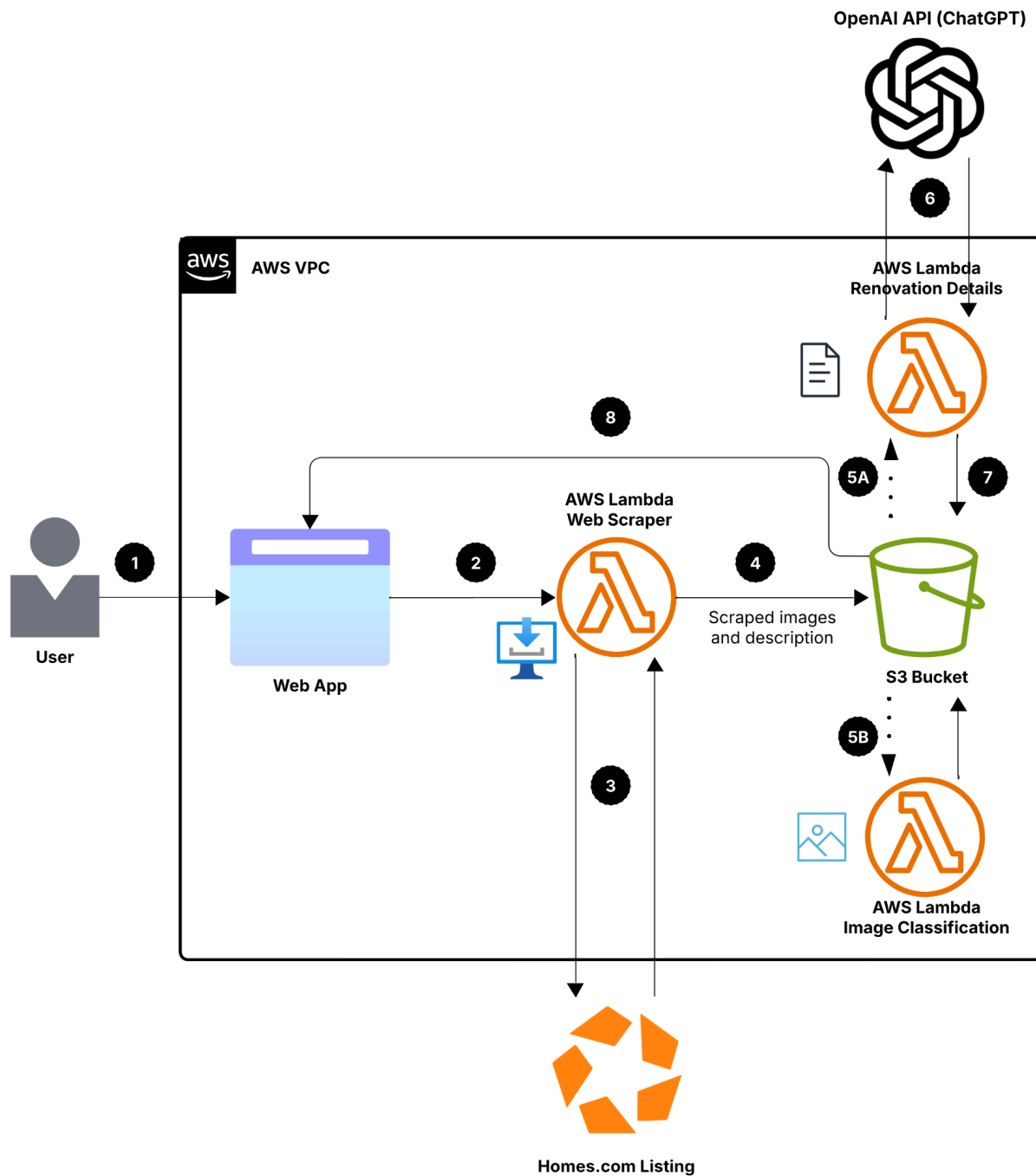
Living Room

Bathroom

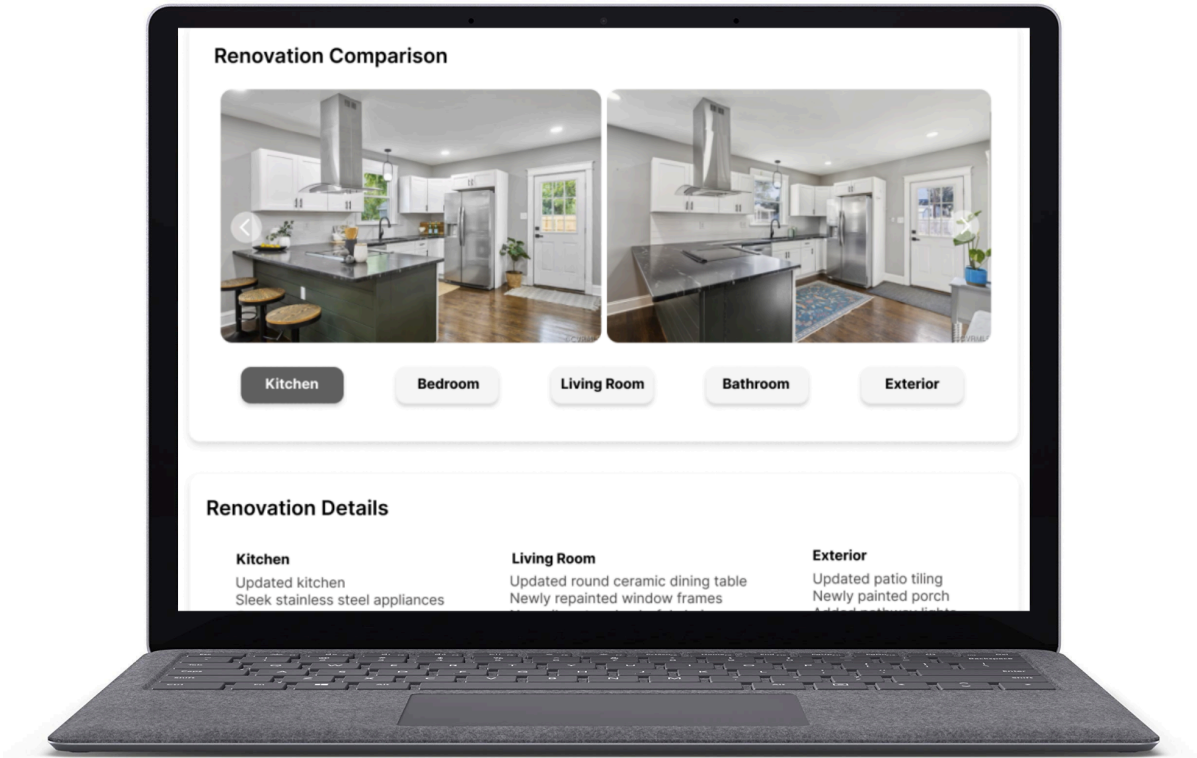
Preliminary Block Diagram for the design of the program: Group Effort 11/5/2024



Architecture flow for the design of the program: Sterling Glasheen 2/30/2025



Second comparison prototype: Courtney Van 3/10/2025



Section E. Concept Evaluation and Selection

To determine the most effective design for the RenovationTracker system, the team used a structured decision-making process supported by a decision matrix. This process allowed the team to compare potential design architectures based on clearly defined criteria and objective metrics.

Selection Criteria and Weighting

The following criteria were used to evaluate each design concept, based on their relevance to the project's goals and client priorities:

- **Performance (45%):** Accuracy and efficiency of renovation detection and room classification.
- **Scalability (25%):** Ability to handle growing volumes of listing data without performance loss.
- **Ease of Integration (20%):** Effort required to connect backend processing with the frontend.
- **Reliability (10%):** System consistency, fault tolerance, and low error rates during testing.

Evaluation Metrics

- **Performance:** Model accuracy (%) and response time per listing.
- **Scalability:** Time to process large datasets (e.g., per 100 listings).
- **Ease of Integration:** Estimated developer hours required to connect components and display results.
- **Reliability:** Percentage of successful outputs during internal testing.

Design Concepts Evaluated

1. **Parallel Lambda Functions (Final Architecture)**
Separate Lambda functions for NLP and image classification, each triggered by S3 uploads and writing output to S3.

2. **Single Lambda Function (Monolithic)**

One Lambda function that performs both NLP and image processing sequentially.

3. **Manual Annotation with Prebuilt Tools**

An approach using AWS Rekognition or similar tools with human oversight for tagging.

Design Concept	Performance (45%)	Scalability (25%)	Ease of Integration (20%)	Reliability (10%)	Weighted Total
Parallel Lambda Functions	9 (4.05)	9 (2.25)	8 (1.6)	8 (0.8)	8.7
Single Lambda Function	7 (3.15)	6 (1.5)	5 (1.0)	6 (0.6)	6.25
Manual Annotation with Prebuilt Tools	5 (2.25)	4 (1.0)	4 (0.8)	7 (0.7)	4.75

Selected Design Concept

The **Parallel Lambda Functions** architecture was selected as the optimal solution. It scored highest across performance, scalability, integration ease, and reliability. This approach aligns with the project's modular goals and ensures each function can be tested and deployed independently.

Rationale for Selection

- Allows NLP and image classification to run independently, enabling parallel processing.
- Simplifies scaling, as each function can handle its own workload triggered by S3 events.
- Reduces integration complexity by isolating tasks and improving modularity.
- Demonstrated reliable output generation across multiple test listings.

Client Feedback

The proposed architecture was shared with the CoStar team, and the feedback confirmed that this approach aligned with the sponsor's expectations for modularity, cloud compatibility, and extensibility.

Next Steps

With the architecture finalized, the team focused on implementing and testing the NLP and image classification Lambda functions, integrating them with the React frontend, and preparing the system for final demonstration and deployment.

Section F. Design Methodology

The RenovationTracker system was developed using an iterative engineering design process, combining computational modeling, experimental validation, and modular system design. This section outlines the methods used to evaluate and evolve the system toward meeting the established design objectives and client needs. Verification was achieved by comparing system performance to project specifications, while validation ensured the system functioned as intended in a real-world context.

F.1 Computational Methods

The RenovationTracker system relies on two core computational models:

- **Natural Language Processing (NLP):**
A language model powered by OpenAI extracts renovation details from property descriptions. The model identifies renovation-relevant phrases and organizes them by room type in a structured JSON format. It was tested against a manually annotated dataset to optimize real estate-specific output.
- **Computer Vision (CV):**
A PyTorch-based image classification model detects room types (e.g., Kitchen, Livingroom) from listing images. The model was trained using a labeled dataset of 500 room images and fine-tuned for common real estate categories. Accuracy was evaluated on a separate validation set.

Both models run as separate AWS Lambda functions, triggered independently by new file uploads to S3. This approach allows for scalable, event-driven execution without manual orchestration. Outputs from both models are written to S3 and later merged client-side by the frontend.

F.2 Experimental Methods

To assess model performance, controlled experiments were conducted using curated datasets:

- **Text Analysis:**
A test set of property descriptions was manually annotated with expected renovation phrases. The model's outputs were evaluated using precision, recall, and F1-score to quantify its effectiveness.

- **Image Classification:**
Room classification accuracy was measured by comparing predicted room types to labeled ground truth data. A confusion matrix was generated to analyze model performance across categories.
- **User Testing:**
Preliminary user testing involved presenting sample outputs (merged renovation and image data) through the React frontend. Feedback was gathered through tests to assess clarity, relevance, and ease of interpretation. Results helped inform updates to output formatting and frontend layout.

F.3 Architecture / High-Level Design

The system follows a modular, cloud-native architecture:

- **Web Scraper:** Collects property descriptions and images from homes.com.
- **NLP Lambda Function:** Extracts renovation details and outputs categorized JSON.
- **Computer Vision Lambda Function:** Classifies room types based on uploaded images.
- **S3 Buckets:** Act as the central communication layer between components.
- **React Frontend:** Combines and displays renovation and image data interactively.

Each component functions independently, which improves scalability and simplifies maintenance. Outputs are merged on the frontend, eliminating the need for server-side orchestration tools like Step Functions.

F.4 Validation Procedure

Validation of the RenovationTracker system was carried out through ongoing collaboration with the project sponsor team at CoStar. Throughout the development process, the team regularly met with CoStar representatives Justin and Abigail to review system behavior, confirm design decisions, and align implementation with client expectations. These working sessions provided valuable feedback on functionality, user experience, and technical feasibility.

Rather than a single validation event, feedback was integrated iteratively during implementation to ensure the system met the intended use case. Design refinements such as output formatting, room classification labels, and frontend layout were made based on this input, ensuring that the final product reflects project goals.

Summary of Design Methodology

The RenovationTracker design methodology employs advanced computational methods, rigorous experimental validation, and an iterative feedback-driven development process. By leveraging these approaches, the team ensured the system met all technical specifications and client requirements while remaining adaptable to future enhancements.

Section G. Results and Design Details

This section presents the results and technical details of the RenovationTracker system. It highlights progress made in model development, system integration, and testing. The following subsections summarize the current performance of core components, preliminary outputs, and design specifications.

G.1 Modeling Results

Text Analysis

The NLP module, powered by OpenAI's language model, successfully extracts renovation details from property descriptions. The system produces structured JSON output with renovations grouped by room when applicable. It performs well even when descriptions are written in casual or inconsistent language.

Image Classification

The computer vision model built with PyTorch classifies room types from property images with 90% test accuracy. It was trained and tested on a dataset of 500 labeled images across common categories (e.g., kitchen, bathroom, living room).

Updated Workflow

The current system architecture uses two separate AWS Lambda functions:

- One handles text extraction using NLP.
- One handles room classification from images.

Each Lambda is triggered by new file uploads to S3, and both write results to the output folder in S3. These outputs are then combined by the frontend application to be then stored in a folder in the S3 bucket as a stored listing

G.2 Experimental Results

Integration Testing

The team verified that the Lambda functions process scraped data from homes.com correctly. Uploads to S3 automatically trigger analysis, and results are accessible within seconds. Both text and image outputs are written to S3 in structured formats.

Error Handling

The system handles missing or unclear input gracefully. For example, if no renovation is detected in a description or the image lacks clear features, the output includes empty arrays without causing failures in downstream processing.

G.3 Prototyping and Testing Results

Web Scraper:

The scraper is successfully pulling text descriptions and images from homes.com. This data serves as input for the Lambda functions, with property details and room images captured with a 95% success rate.

Prototype Outputs:

Image Classification Output:

```
{  
  "filename": "liv2.jpg",  
  "roomType": "Livingroom",  
  "score": 83  
}
```

Text Analysis Output:

```
{  
  "Bedroom": [],  
  "Bathroom": [  
    "Updated fixture and mirror",  
    "Updated vanity, light fixture, and mirror"  
  ],  
  "Kitchen": [  
    "Stainless appliances",  
    "Mobile island with breakfast bar"  
  ],  
  "Livingroom": [  
    "New modern light fixture"  
  ],  
  "Exterior": [  
    "Recently re-stained porch",  
    "Fresh paint throughout"  
  ]  
}
```

Unified Listing Format (Used by Frontend):

```
{  
  "address": "702 Mount Vernon Ave, Alexandria VA",  
  "timestamp": "2025-03-09T22:40:00Z",  
  "images": {  
    "Bedroom": {  
      "historical": "Histbed.jpg",  
      "current": "bed.jpg"  
    },  
    "Bathroom": {  
      "historical": "Histtoil.jpg",  
      "current": "toil.jpg"  
    },  
    "Kitchen": {  
      "historical": "Histkit.jpg",  
      "current": "kit.jpg"  
    },  
    "Livingroom": {  
      "historical": "Histliv.jpg",  
      "current": "liv.jpg"  
    },  
    "Exterior": {  
      "historical": "Histdoor.jpg",
```

```
    "current": "door.jpg"
  }
},
"renovationDetails": {
  "Bedroom": [],
  "Bathroom": [
    "Updated fixture and mirror",
    "Updated vanity, light fixture, and mirror"
  ],
  "Kitchen": [
    "Stainless appliances",
    "Mobile island with breakfast bar"
  ],
  "Livingroom": [
    "New modern light fixture"
  ],
  "Exterior": [
    "Recently re-stained porch",
    "Fresh paint throughout"
  ]
}
```

G.4 Final Design Details and Specifications

System Architecture

The system is composed of:

- A web scraper that collects listings.
- Two independent Lambda functions for NLP and image classification.
- S3 buckets that store inputs and outputs.
- A React frontend that reads from S3 and presents merged renovation data.

Scalability

The architecture supports horizontal scaling. Multiple Lambda functions can run in parallel, and S3 ensures high availability and throughput.

Design Specifications

- NLP module processes property descriptions with approximately 85% useful extraction rate.
- Image classification model maintains 90% accuracy on test data.
- Outputs are stored in JSON format and support frontend merging by room type.

Next Steps

- Expand the dataset to include more diverse property types and listing formats.
- Collect end-user feedback for further UI and UX improvements.

These results demonstrate that the RenovationTracker system meets key design requirements and is progressing toward a robust, scalable solution for real estate renovation detection.

Section H. Societal Impacts of Design

Beyond meeting technical specifications, the RenovationTracker project was developed with an awareness of the broader impacts it may have on individuals, the real estate industry, and society at large. As a tool designed to improve how renovation information is communicated, its influence extends to areas such as public trust in listings, regulatory compliance, property values, and sustainable renovation practices.

This section explores how the design addresses considerations related to public welfare, social communication, economic opportunity, environmental responsibility, and ethical data usage. By incorporating these perspectives throughout development, the team aimed to create a system that is not only functional, but also responsible, inclusive, and aligned with long-term industry and societal needs.

H.1 Public Health, Safety, and Welfare

The RenovationTracker design prioritizes accuracy and reliability to ensure the welfare of its users, including buyers, sellers, and real estate professionals. By automating the renovation detection and room classification processes, we eliminate human error and reduce the risks of misrepresentation in property listings, which can lead to financial or emotional distress for buyers and sellers. Additionally, the interactive visualization features provide users with a clear and comprehensive understanding of renovations, enabling more informed decisions and enhancing overall consumer trust. The system adheres to data privacy and security standards to protect sensitive user data and maintain ethical data usage practices.

H.2 Societal Impacts

RenovationTracker has the potential to transform how property renovations are perceived and evaluated, creating a more transparent and engaging property marketplace. By bridging the gap between static descriptions and dynamic visualization, the tool fosters better communication and understanding between buyers and sellers, ultimately strengthening trust in real estate transactions. This transparency can also inspire homeowners to invest in quality renovations, knowing their efforts will be clearly represented and appreciated in the market.

H.3 Political/Regulatory Impacts

The design operates within the regulatory frameworks of data protection laws, such as GDPR and CCPA, ensuring compliance with privacy standards in different regions. The ability to provide transparent and accurate property data could also encourage the adoption of stricter guidelines for property listings, promoting fair practices across the industry. Furthermore, RenovationTracker's automation of renovation tracking could influence regulatory discussions on the standardization of property data in real estate.

H.4. Economic Impacts

RenovationTracker has the potential to increase the value of properties by clearly showcasing renovations, resulting in higher selling prices and quicker sales. For buyers, the tool helps identify properties with higher value, reducing the risk of overspending on homes with unverified updates. Real estate agencies may also benefit economically from increased user engagement, faster property transactions, and improved client satisfaction. On a broader scale, this tool could contribute to market efficiency by better connecting buyers and sellers through more accurate and visually compelling listings.

H.5 Environmental Impacts

While RenovationTracker primarily focuses on property renovations, it indirectly encourages environmentally friendly updates by emphasizing the value of sustainable features such as energy-efficient appliances and eco-friendly materials. By promoting these updates in property listings, the design could influence market trends toward greener renovations. However, we acknowledge the environmental cost of running machine learning models and are committed to optimizing computational efficiency to minimize energy consumption.

H.6 Global Impacts

The global real estate market could benefit from RenovationTracker by offering a unified platform that standardizes renovation documentation and presentation across regions. The scalability of the system allows for seamless integration in various countries, potentially influencing global real estate practices. Additionally, its multilingual capabilities could make property listings more accessible to international buyers, fostering cross-border real estate investments and transactions.

H.7. Ethical Considerations

The ethical design of RenovationTracker prioritizes transparency, fairness, and accuracy. By ensuring that renovations are represented truthfully, we aim to prevent the misrepresentation of property values and reduce instances of fraud. The project also adheres to ethical AI development principles, including minimizing bias in NLP and computer vision models, ensuring inclusivity in property data, and maintaining user privacy. Future updates may include user feedback mechanisms to address inaccuracies and ensure continuous improvement.

Section I. Cost Analysis

The development of RenovationTracker required minimal direct expenditures. The primary cost incurred was approximately \$7 for usage of OpenAI's API during the testing and refinement of renovation related outputs. All other tooling and platforms, including development environments and libraries, were free or open source.

Operational costs related to AWS services such as Lambda functions and S3 storage were covered through credits provided by our project sponsor, Syed, who coordinated with AWS to secure the necessary cloud resources. This support allowed the team to run and test the system at scale without introducing additional financial strain.

No other licensing, infrastructure, or third-party service fees were required. All major project tasks were completed using either sponsor-supported or no cost solutions. Given the project's software based nature and remote-friendly development, no physical materials or hardware were needed.

Overall, the RenovationTracker project remained well within budget, with no significant costs beyond cloud services and API usage.

Section J. Conclusions and Recommendations

The RenovationTracker project evolved through a focused and collaborative design process guided by our sponsor, Syed, and supported by CoStar team members Justin and Abigail. From the beginning, the project was structured around four key areas: web scraping, prompt engineering, image recognition, and front-end integration. This modular approach allowed each team member to take ownership of a component while contributing to a unified system.

Syed's structured guidance helped the team break down complex tasks and maintain progress. Justin and Abbi played an essential role throughout development by offering real estate expertise, resolving technical issues, and ensuring alignment with CoStar's standards. Their availability and feedback helped us address challenges quickly and stay on track.

The final system includes:

- A **functional web scraper** that collects listing descriptions and images from homes.com.
- An **OpenAI-powered NLP function** that extracts and organizes renovation details into a structured format.
- A **PyTorch-based image classification model** that labels room types with high accuracy.
- A **React frontend** that merges and displays renovation and room classification outputs in a clear, interactive format.

Throughout the design process, we prioritized usability, modularity, and scalability. One major change was the decision to replace AWS Step Functions with a simplified and more effective approach using separate Lambda functions and client-side merging. This shift improved maintainability and allowed for faster iteration.

Lessons Learned:

- Dividing the project into clear areas of responsibility improved productivity and integration efficiency.
- Real-time support from CoStar advisors accelerated problem-solving and reduced downtime.
- Staying focused on real-world application ensured the tool remained practical and client-aligned.

Recommendations and Future Work:

While RenovationTracker is functional, several areas offer potential for future improvement:

- **Improved Room Differentiation:** Enhance the image classification pipeline to distinguish between multiple rooms of the same type (e.g., identifying "Bathroom 1" vs. "Bathroom 2") to avoid mismatches when pairing renovation descriptions with images.
- **Renovation-Rich Listings:** Investigate ways to improve the quality of scraped listings from homes.com, including filtering for renovation-heavy descriptions or exploring partnerships to access more detailed listing data.
- **User Feedback Integration:** Add a simple user feedback mechanism to allow corrections to misclassified rooms or renovation text, enabling continuous system refinement.
- **Database Storage and Search:** Integrate a backend database to store listing results, enabling persistent storage, advanced filtering, and search capabilities for users.

In conclusion, RenovationTracker successfully delivers a working system that enhances how property renovations are presented. The team overcame several technical hurdles, refined its architecture, and met the project's core goals. The system is scalable, accurate, and adaptable and with continued support and iteration, it has the potential to significantly impact how renovation data is visualized and understood in the real estate market.

Renovation visualization tool development	3 weeks	February 14, 2025
Final system integration and refinement	3 weeks	March 20, 2025
Final testing and quality assurance	2 weeks	April 3, 2025
Final report and Capstone EXPO preparation	3 weeks	April 24, 2025
Capstone EXPO presentation	1 week	April 25, 2025

Appendix 2: Team Contract (i.e. Team Organization)

Copy and paste the content from the completed Team Contract here starting with Step 1 of the Team Contract and including all content following the 'Contents' list.

Step 1: Get to Know One Another. Gather Basic Information.

Task: This initial time together is important to form a strong team dynamic and get to know each other more as people outside of class time. Consider ways to develop positive working relationships with others, while remaining open and personal. Learn each other's strengths and discuss good/bad team experiences. This is also a good opportunity to start to better understand each other's communication and working styles.

<i>Team Member Name</i>	<i>Strengths each member bring to the group</i>	<i>Other Info</i>	<i>Contact Info</i>
Sterling Glasheen	Communication, problem solving, open-minded, adjustable,	Open to working to lead the team to achieving a great working product.	glasheensm@vcu.edu (804) 971-2901
Jaidon Lightfoot	Communication, adaptability, problem solving, open-minded	Works as a receiving assistant in the ERB.	lightfootjb@vcu.edu (757)663-1692 lightfxxt on discord
Courtney Van	Communication, organization, problem-solving	Currently works in Open Cyber City Lab as a Research Assistant, working with machine learning in Python	vanct2@vcu.edu 572-383-6353 cornbread on discord
Jonathan Netala	Communication, attention-to-detail, problem-solving, resourcefulness	Experience working with machine learning models and automating data processes for real-world applications; in the past, e.g. a financial market analysis tool	netalajd@vcu.edu (804)-385-9408 pakalutupaki on discord

<i>Other Stakeholders</i>	<i>Notes</i>	<i>Contact Info</i>
Preetam Ghosh	<i>No notes</i>	pghosh@vcu.edu
Syed Khajamoinuddin	<i>No notes</i>	ksyed@costar.com

Step 2: Team Culture. Clarify the Group's Purpose and Culture Goals.

Task: Discuss how each team member wants to be treated to encourage them to make valuable contributions to the group and how each team member would like to feel recognized for their efforts. Discuss how the team will foster an environment where each team member feels they are accountable for their actions and the way they contribute to the project. These are your Culture Goals (left column). How do the students demonstrate these culture goals? These are your Actions (middle column). Finally, how do students deviate from the team's culture goals? What are ways that other team members can notice when that culture goal is no longer being honored in team dynamics? These are your Warning Signs (right column).

Resources: More information and an example Team Culture can be found in the Biodesign Student Guide "Intentional Teamwork" page ([webpage](#) | [PDF](#))

<i>Culture Goals</i>	<i>Actions</i>	<i>Warning Signs</i>
Clearly defined goals	- track progress - setting realistic goals - roles are clear	- Student missed deadlines - student overwhelmed - task is not complete
Open communication	- promote accountability - responding in a timely manner - open to criticism	- misunderstanding/confusion - not speaking up
Have fun	- Celebrate wins - encourage each other - recognize achievements	- overly stressed - unenthusiastic meetings - low energy

Step 3: Time Commitments, Meeting Structure, and Communication

Task: Discuss the anticipated time commitments for the group project. Consider the following questions (don't answer these questions in the box below):

- What are reasonable time commitments for everyone to invest in this project?
- What other activities and commitments do group members have in their lives?
- How will we communicate with each other?
- When will we meet as a team? Where will we meet? How Often?
- Who will run the meetings? Will there be an assigned team leader or scribe? Does that position rotate or will same person take on that role for the duration of the project?

Required: How often you will meet with your faculty advisor, where you will meet, and how the meetings will be conducted. Who arranges these meetings?

See examples below.

<i>Meeting Participants</i>	<i>Frequency Dates and Times / Locations</i>	<i>Meeting Goals Responsible Party</i>
Students Only	As Needed, On Discord Voice Channel or in text channels	Update group on day-to-day challenges and accomplishments and relay what needs to be done
Students Only	Every Thursday during class time, on discord	Actively work on the project. Discuss ideas and plan for the future. Assign tasks.
<i>Students + Faculty advisor</i>	3:30 Every Friday on Microsoft Teams	Updates on tasks and assigning of tasks
<i>Project Sponsor</i>	3:30 Every Friday on Microsoft Teams and 1-3 every Thursday in person at the CoStar Office.	Updates on tasks and assigning of tasks

Step 4: Determine Individual Roles and Responsibilities

Task: As part of the Capstone Team experience, each member will take on a leadership role, *in addition to* contributing to the overall weekly action items for the project. Some common leadership roles for Capstone projects are listed below. Other roles may be assigned with approval of your faculty advisor as deemed fit for the project. For the entirety of the project, you should communicate progress to your advisor specifically with regard to your role.

- **Before meeting with your team**, take some time to ask yourself: what is my “natural” role in this group (strengths)? How can I use this experience to help me grow and develop more?
- **As a group**, discuss the various tasks needed for the project and role preferences. Then assign roles in the table on the next page. Try to create a team dynamic that is fair and equitable, while promoting the strengths of each member.

Communication Leaders

Suggested: Assign a team member to be the primary contact for the client/sponsor. This person will schedule meetings, send updates, and ensure deliverables are met.

Suggested: Assign a team member to be the primary contact for faculty advisor. This person will schedule meetings, send updates, and ensure deliverables are met.

Common Leadership Roles for Capstone

1. **Project Manager:** Manages all tasks; develops overall schedule for project; writes agendas and runs meetings; reviews and monitors individual action items; creates an environment where team members are respected, take risks and feel safe expressing their ideas.
Required: On Edusourced, under the Team tab, make sure that this student is assigned the Project Manager role. This is required so that Capstone program staff can easily identify a single contact person, especially for items like Purchasing and Receiving project supplies.
2. **Logistics Manager:** coordinates all internal and external interactions; lead in establishing contact within and outside of organization, following up on communication of commitments, obtaining information for the team; documents meeting minutes; manages facility and resource usage.
3. **Financial Manager:** researches/benchmarks technical purchases and acquisitions; conducts pricing analysis and budget justifications on proposed purchases; carries out team purchase requests; monitors team budget.
4. **Systems Engineer:** analyzes Client initial design specification and leads establishment of product specifications; monitors, coordinates and manages integration of sub-systems in the prototype; develops and recommends system architecture and manages product interfaces.
5. **Test Engineer:** oversees experimental design, test plan, procedures and data analysis; acquires data acquisition equipment and any necessary software; establishes test protocols and schedules; oversees statistical analysis of results; leads presentation of experimental finding and resulting recommendations.

